28	To observe diffraction rings by a carbon film, a beam of electrons is accelerated from rest across a potential difference of *V* so that the de Broglie wavelength of the electrons is 1.0 x 10^{-10} m. What is the value of *V* ?							
	A	90 V	**B**	150 V	**C**	270 V	**D**	330 V
L2	**Answer: B** de Broglie wavelength, $p = \frac{h}{\lambda}$ ------ (1) By conservation of energy, Loss in electric potential energy = Gain in kinetic energy							

$qV = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ ------ (2)

Sub (1) into (2),

$$qV = \frac{h^2}{2m\lambda^2}$$

$$V = \frac{h^2}{2qm\lambda^2} = \frac{(6.63\ x\ 10^{-34})^2}{2(1.60\ x\ 10^{-19})(9.11\ x\ 10^{-31})(1.0\ x\ 10^{-10})^2}$$

$V = 150\ V$ (2 s.f.)

29 A nuclear fusion reaction is as follows: